

# Warrant

## Complete system reference

An estate settlement decision, investigation and filing engine in which every fact carries the sentence that justifies it, every rule carries its statute, and no model runs in the decision path.

### Build state

171 TypeScript tests · 11 Rust tests · clean production build · thirteen interface panes

**Verified:** all three governing California thresholds confirmed against Judicial Council form DE-300 [Rev. April 28, 2025]

**Not yet done:** no live model call, no live Anvil call, extraction stubbed, 55 of 58 counties unresearched

Built for the Alix “Agents of Administration” hackathon

Track 1 — Rules as Data, with a Track 3 integration

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# 1. The problem

An executor is handed the largest administrative job of their life at the worst possible moment. They must work out which of six California probate procedures applies, assemble a complete picture of what the deceased owned across institutions that will not talk to them without a court order, file the right forms with the right county in the right sequence, and avoid distributing a penny before it is safe to do so. Most people do it once, badly, while holding down a job.

Alix exists to do that work for ordinary families at a price ordinary families can pay, by replacing the twelve-person private-bank back office with agents. The obstacle is not capability. A model can read a bank statement. The obstacle is **trust**: an estate decision that is wrong costs the family six months, and no settlement specialist will hand work to a system that cannot show where its answer came from.

## 1.1 What the mentors actually said

*“Assume the LLM will be wrong one out of ten times. How do you build a system around assuming it's going to be wrong?”*

— Soren, Rules as Data track

*“Dear LLM, when you make a claim, you have to give me the verbatim quote. And then I have a deterministic system actually go in, check the transcript, and say, I need that exact quote to be in there. And if it's not, then I can't trust this claim.”*

— Soren

*“Tomorrow, if your system told me that I have to go through formal probate, you need to track me every decision that was made. You need to tell me where you got your legal information.”*

— Soren

*“The two things I always think about are: how reversible is it, and what's the blast radius? If I'm telling you that you should email your client, I don't care if the LLM is wrong. If I'm telling you that you need to file this formal probate form and you're wrong, it's another six months.”*

— Soren

*“A fun problem is, how do you know what you don't have? The transcript mentioned the car. But I don't know the value of the car. I don't know the license plate.”*

— Soren

*“There are three thousand counties in America. Each county has their own rules for probate. You can't just have a Python file for each.”*

— Soren

*“We need a universal document store that captures all the relevant details that make it unique — not just the form ID that's written on the PDF, but where did it come from, who are the relevant parties, what the jurisdiction is.”*

— Ian, Paperwork Killed track

*“If someone has to go in and every week write a weekly email to someone, that's a lot of work. If I can AI generate the email, all I do as a specialist is review that email and send it.”*

— Hugh, Alix

*“Even the best constructed trust inevitably leaves things dangling.”*

— Hugh

Every one of those statements is answered by a named component in this system. Section 3 maps them one to one.

## 1.2 The one number that proves the point

California's small-estate thresholds moved on 1 April 2025 under AB 2016. The previous figures — \$184,500 and \$61,500 — remain on a great many web pages, including some of California's own self-help material. Every language model's training data contains them.

### **A council of five models would have agreed, confidently, and been wrong.**

This is why debate cannot substitute for source verification. On the single most important number in the system, consensus reproduces the error. Only fetching the Judicial Council's own form settles it — which is what was done, and the correct figures are \$208,850 and \$69,625, with the primary-residence cap at \$750,000.

## 2. Architecture

Two rules govern the entire design. Everything else is consequence.

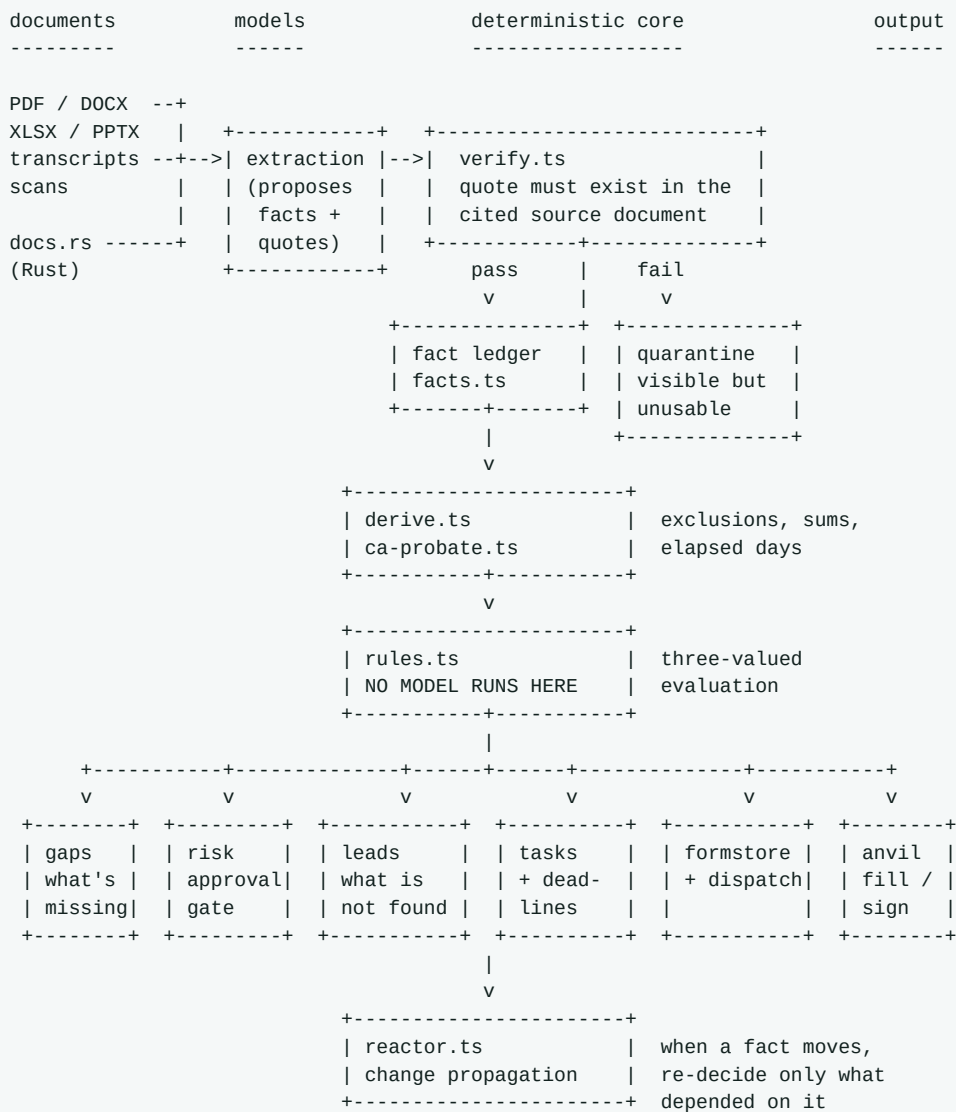
### 1. Models propose, code decides.

No LLM call sits anywhere in the decision path. Models turn prose into candidate facts. Deterministic code does the arithmetic, applies the rules, and produces the answer.

### 2. Nothing enters a decision without a warrant.

A fact needs either a verbatim quotation located in a source document, or a derivation naming the facts it came from. There is no third option.

### 2.1 Data flow



### 2.2 Technology

| Layer     | Choice                                  | Why                                                                  |
|-----------|-----------------------------------------|----------------------------------------------------------------------|
| Shell     | Tauri 2                                 | Desktop app; secrets stay in a Rust process the renderer cannot read |
| Engine    | TypeScript, pure functions              | The decision layer is testable offline with no network and no keys   |
| Ingestion | Rust — pdf-extract, calamine, quick-xml | Binary parsing where it belongs, with real fixtures under test       |
| Interface | React 19, Tailwind 4                    | Thirteen panes, all reading precomputed state                        |
| Forms     | Anvil REST + GraphQL                    | Register, reconcile, fill, sign, retrieve                            |
| Models    | Five providers, IDs as data             | Vendors rename models constantly; a rename is a config edit          |

The engine is deliberately independent of the shell. Every module below is a pure function over facts, which is why the whole demonstration runs with no network call and cannot fail on a venue's wireless.

### 3. Answering the brief, point by point

| What was asked for                               | Component                                 | Section  |
|--------------------------------------------------|-------------------------------------------|----------|
| Assume the model is wrong 10% of the time        | Fact ledger + quarantine                  | 4.3      |
| Verbatim quote, checked deterministically        | verify.ts                                 | 4.2      |
| Track every decision and where the law came from | Authority on every rule; evaluation trace | 5.2      |
| Reversibility and blast radius                   | risk.ts approval gate                     | 5.6      |
| How do you know what you don't have?             | Three-valued logic; gaps.ts; leads.ts     | 5.3, 6.1 |
| How do you determine when this runs?             | reactor.ts dependency tracking            | 5.5      |
| 3,000 counties, no Python file per county        | Rules as data; county registry            | 5.2, 8.2 |
| Universal document store                         | formstore.ts composite identity           | 7.1      |
| Map our data onto the PDF                        | anvil.ts field bindings                   | 7.3      |
| Signature at spot X and Y                        | SignatureField rects in PDF points        | 7.1      |
| Dispatch — email, fax, post, print               | planDispatch                              | 7.2      |
| Generate the weekly email; specialist reviews    | correspondence.ts                         | 9.1      |
| Is this account the one from the will?           | entities.ts                               | 9.2      |
| Take hours out of the system                     | effort.ts, measured honestly              | 9.3      |



## 4. Evidence layer

### 4.1 Document ingestion

Rust module `docs.rs`. Extracts text from PDF, XLSX, XLSM, XLS, DOCX, PPTX and plain text.

- Spreadsheets retain sheet and row structure rather than collapsing to a blob.
- Word runs split mid-sentence are rejoined. This is load-bearing rather than cosmetic: a quote verifier fails against a document whose sentences the extractor shredded, so extraction quality directly determines how much evidence survives.
- Malformed PDFs are contained rather than crashing the run.
- Low-yield PDFs are flagged as probable scans rather than imported blank, so a scanned document is a known gap instead of a silently empty one.

Tested against committed real Office and PDF fixtures, so a checkout that mangles binaries fails loudly rather than at demonstration time.

### 4.2 Deterministic quote verification

Module `verify.ts`. Given a claimed quotation and a document set, locates it or proves it absent.

- Normalised exact match first, tolerating whitespace differences and smart quotes.
- Then n-gram coverage, so paraphrase degrades smoothly below verbatim instead of passing or failing on a knife edge.
- Returns a character span, so the interface can open the source and highlight the sentence.

| Verdict     | Meaning                                                  | Admitted? |
|-------------|----------------------------------------------------------|-----------|
| verified    | Exact quotation located in the cited document            | Yes       |
| loose       | Found in a different document, or only partially matched | No        |
| unsupported | Does not appear anywhere in the data room                | No        |
| no_citation | The model supplied no usable quotation                   | No        |

#### No model participates in verification.

We never ask a second model whether the first was honest. The check is string matching against the source, so fabrication is caught with certainty rather than probability. This is the single most important design decision in the system, and it is the one Soren named as the crux of his track.

### 4.3 The fact ledger

Module `facts.ts`. Every fact carries a warrant, of one of two kinds.

```
interface QuoteWarrant {
  kind: "quote";
  document: string;           // the document the model cited
  quote: string;              // the verbatim sentence it relied on
  verdict: Verdict;           // from verify.ts
  similarity: number;         // 0-1, for diagnostics
  span?: { start, end };      // character offsets, for click-through
  matchedDocument?: string;   // where it was really found, if elsewhere
}

interface DerivedWarrant {
  kind: "derivation";
  formula: string;            // human-readable
  inputs: FactKey[];          // every fact consumed
  authority?: { citation, sourceUrl };
}
```

Facts also carry an asOf date (a balance is only true on a date), supersession, and the model that proposed them.

**Only exact quotation earns admission.** A loose match is deliberately not good enough. In a domain where being wrong costs a family six months, “probably said something like this” is not evidence.

ledger() returns verified facts only, so a quarantined fact is **structurally unreachable** by the rules engine — not filtered late, but absent from the data structure the engine reads. It remains visible in the interface so a reviewer can see exactly what the model tried to assert and on what basis.

integrity() scores only model-proposed facts. Derived totals are excluded, because counting them would flatter the number.

## 5. Reasoning layer

### 5.1 Derivation

Module `derive.ts`, with California specifics in `ca-probate.ts`. No model performs arithmetic anywhere in this system. Models extract individually-quoted values; code sums them under a cited rule.

- Statutory exclusions are data: joint tenancy, named beneficiary, funded trust, registered vehicle, property passing to a surviving spouse — each with its citation.
- Asset facts follow `asset.<id>.value` and `asset.<id>.<flag>`, so a new asset class needs no code change, no schema migration and no deploy.
- Elapsed-time facts recompute every run, so a statutory waiting period expiring is itself a change the reactor notices.

The derived total records its formula and every input key, so a sum is as auditable as the quotations beneath it.

### 5.2 Rules as data

Module `rules.ts`. A rule is a JSON-serialisable object.

| Field                                                   | Purpose                                                                        |
|---------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>decisionPoint</code>                              | Groups rules competing to answer the same question                             |
| <code>jurisdiction</code>                               | State, optionally county — a county rule overlays statewide automatically      |
| <code>requires</code>                                   | Facts that must be held before the rule can be evaluated at all                |
| <code>when</code>                                       | Predicate tree: all, any, not, {fact, op, value}, exists, missing              |
| <code>then</code>                                       | Conclusion, forms, obligations, timeline range, cost range                     |
| <code>authority</code>                                  | Citation, source URL, effective date, and the date we read the source          |
| <code>estimates</code>                                  | Names any figure in <i>then</i> that is a practice estimate, not a sourced one |
| <code>priority</code>                                   | Resolves competition when several rules fire at one decision point             |
| <code>blastRadius,</code><br><code>reversibility</code> | Drive the human-approval gate                                                  |

#### Provenance decays, so record when you looked.

*retrievedAt* is as important as the citation. California's thresholds moved in April 2025 and most of the internet has not noticed. A rule that cannot say when it was last checked cannot be trusted a year from now.

#### An uncited number beside cited ones is the worst possible error.

So *estimates* is mandatory rather than optional. Where a timeline or a cost is a practice estimate rather than a statutory figure, the rule says so and the interface renders the distinction.

### 5.3 Three-valued evaluation

A predicate over a fact we do not hold evaluates to **unknown**, never to false. Kleene logic in *and*, *or* and *not*.

```
all:  any false -> false;  else any unknown -> unknown;  else true
any:  any true  -> true;   else any unknown -> unknown;  else false
not:  unknown stays unknown
```

A rule whose condition is unknown does not quietly fail to fire. It reports itself **blocked** and names the fact keys it needed. Gap detection therefore falls out of the same machinery that decides, rather than being a separate feature that could drift out of agreement with it.

Evaluation records every fact key consulted, and produces a step-by-step trace rendered in the Decisions pane so a reviewer can follow the reasoning line by line.

## 5.4 Decision resolution

Rules are grouped by decision point and scoped by jurisdiction. Within a point, fired rules resolve on priority; losers are retained as *alsoFired* so the choice stays auditable. Blocked and not-applicable rules are reported separately rather than discarded.

## 5.5 Dependency-tracked re-evaluation

Module `reactor.ts`. This answers Soren's question about when the system should run.

- Diffs two ledgers by decision-visible value.
- Intersects the changed fact keys with each decision's recorded dependency set.
- Re-runs only the affected decision points and carries the rest forward unchanged.
- Emits a Change Report: the fact that moved, the quotation that moved it, the rules that consequently fired, forms added and removed, obligations gained, and the decision points that were *provably* untouched.

A blocked decision is not automatically re-run. The evaluator records a key even when that key was absent, so a missing fact is already part of the dependency set and the intersection test alone is sound. Treating “blocked” as “always stale” would forfeit the skip guarantee for nothing.

### Why this matters commercially.

An estate runs nine to eighteen months and documents arrive throughout. Today each arrival triggers either an expensive full re-review or nothing at all — and “nothing” is how a twenty-thousand-dollar appraisal revision silently invalidates a filing strategy. Bounding re-decision to the dependency closure changes which errors are possible, not merely how fast they are caught.

## 5.6 The approval gate

Module `risk.ts`. Soren's two axes, encoded on every rule.

| Outcome | Condition                                                 | Example                            |
|---------|-----------------------------------------------------------|------------------------------------|
| auto    | Reversible and blast radius not high                      | Transfer a vehicle at the DMV      |
| review  | Irreversible, costly, or high blast radius                | File a petition for formal probate |
| blocked | Cannot conclude, or a supporting fact failed verification | No appraisal exists yet            |

The gate also re-checks that no supporting fact is unverified. The ledger already guarantees this, so the check should never fire — and if it ever does, something upstream is broken and refusing is the correct response.

This is also where Alix's product stance is drawn in code. The service stays human-led; the gate is precisely the boundary at which a specialist's judgement is spent, on the decisions that warrant it rather than on the ones a machine can safely take.

## 6. Investigation layer

### 6.1 The lead engine

Module `Leads`. `ts`. A fact ledger records what documents say. An investigation reasons about what they **imply**.

A wire to Geneva is not an asset. It is evidence that an account exists which no document in our possession describes. A Form 5471 is not a company; it is proof that a company exists somewhere. Making that inference explicit — with the request that would confirm or kill it — is the difference between reading documents and conducting an investigation.

Seventeen patterns, each carrying a rationale, a priority, and the specific actions that would resolve it.

| Group              | Patterns                                                                                                                                     |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Tax-derived        | Schedule B foreign-account answer · Form 8938 · Form 3520 / 3520-A · Form 5471 · Form 8865 · Form 8621 · Form 1116 · foreign pension income  |
| Banking            | International wire · safe-deposit box · currency exchange                                                                                    |
| Life and documents | Foreign passport, address or residency · business interests and K-1s · cryptocurrency · named professionals · employers · unclaimed property |

Each action records who it goes to, the channel, **which authority documents it requires** (certified Letters, death certificate, Form 56, apostille, certified translation, court order), the form where one exists, and what a response may reveal.

Detection reuses the rules engine's predicate evaluator, so patterns share the three-valued semantics rather than growing a second, subtly different evaluator. A pattern that does not fire is reported **dormant**, with the fact key that would trigger it.

#### The dormant list is the executor's defence.

If a beneficiary later argues that the offshore account should have been found, the record shows the pattern was evaluated, what it was waiting for, and that nothing in the data room triggered it.

### 6.2 The administration task graph

Module `tasks`. `ts`. Six phases with enforced dependencies.

| Phase               | Covers                                                                                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Establish authority | Petition and certified Letters, estate EIN, estate bank account, Form 56                                                                                            |
| Secure              | Redirect mail for a full annual cycle; secure property, vehicles and valuables; maintain insurance; preserve devices without accessing them; stop recurring charges |
| Investigate         | IRS transcripts, institution-wide bank searches, public records, master ledger and investigation log                                                                |
| Report              | Final personal return, estate return above \$600 gross income, international review                                                                                 |
| Creditors           | Notice to known and reasonably ascertainable creditors, claim period                                                                                                |
| Close               | Inventory and appraisal, reserves, final distribution, discharge                                                                                                    |

Dependencies are real, not decorative: no bank will speak to you before Letters issue, so bank searches are blocked until the petition completes, and the interface says which task is holding them.

Tasks carry cautions where a wrong move is expensive — do not wipe devices, do not commingle estate funds with your own, and do not file delinquent foreign-account forms before counsel has reviewed whether prior omissions were innocent, negligent or wilful.

### 6.3 Statutory deadlines

Computed from dates in the ledger, never from the system clock. The as-of date is a parameter, so a run is reproducible.

| Deadline                | Period                                                 | Authority         |
|-------------------------|--------------------------------------------------------|-------------------|
| Lodge the will          | 30 days from death                                     | Prob. Code § 8200 |
| Inventory and Appraisal | 4 months from Letters                                  | Prob. Code § 8800 |
| Creditor claims         | Later of 4 months from Letters, or 60 days from notice | Prob. Code § 9100 |
| First publication       | At least 15 days before the hearing                    | Prob. Code § 8121 |
| Mailed notice           | At least 15 days before the hearing                    | Prob. Code § 8110 |
| Federal estate tax      | 9 months from death                                    | IRC § 6075(a)     |
| FBAR                    | Where foreign aggregate exceeded \$10,000              | 31 CFR 1010.350   |

Periods running from the later of two anchors compute both and report which governed.

#### **A deadline whose anchor is missing reports unknown — never “today”.**

A missed statutory deadline is exactly the irreversible, high-blast-radius harm this system exists to prevent, so silence or a guess is not an acceptable output. In the demonstration estate, Letters have not issued, and four deadlines honestly report that they cannot be computed and name the fact they need.

### 6.4 The distribution gate

Distribution is the one irreversible act in an estate. Once money reaches beneficiaries it is practically unrecoverable, and early distribution is the most common route by which an executor becomes personally liable. The gate is therefore deliberately conservative and is held by any of:

- An open **critical** lead — assets may exist that no document describes.
- An unclosed creditor claim period.
- Any fact still required by a pending decision.

## 7. Filing layer

### 7.1 The universal document store

**The identifier printed on a PDF is not unique.**

DE-111 means something only within the California Judicial Council. PRO 010 exists only in Los Angeles. Almost every brokerage prints its own Affidavit of Domicile. Two counties can use the same number for different paper, and the same paper changes when a revision is issued.

Module `formstore.ts`. A record is keyed on **issuer + jurisdiction + revision**, and carries everything an operator needs in order to actually use the form.

| Attribute   | Contents                                                                           |
|-------------|------------------------------------------------------------------------------------|
| Identity    | Composite key, printed identifier, title, issuer, jurisdiction, revision           |
| Parties     | Preparer, signer, notary, witness, recipient — each optionally bound to a fact key |
| Signatures  | Type, page number, and rectangle in PDF points                                     |
| Delivery    | Accepted channels, named recipient, operator notes                                 |
| Constraints | Requires notary, requires wet original                                             |
| Provenance  | Source URL and retrieval date                                                      |
| Binding     | Field bindings, and the Anvil cast identifier once registered                      |

Five issuer kinds are modelled: Judicial Council, county court, state agency, federal agency, and financial institution.

`resolveForm()` **returns ambiguity rather than guessing**. Asking for “Affidavit of Domicile” without naming the institution yields both issuers and a reason. Asking for PRO 010 in San Mateo yields nothing and explains why. Sending the wrong institution's affidavit is a returned filing and another round trip for a grieving family, so a confident wrong answer is worse than no answer.

### 7.2 Dispatch

A filled PDF sitting on a disk has settled nothing. `planDispatch()` chooses a channel from those the recipient accepts, preferring cheapest and fastest — but two conditions override preference entirely.

- A form requiring a wet original or notarisation cannot be delivered electronically, whatever the recipient nominally accepts.
- An institution that requires a telephone call first must be called first. JPMorgan Chase's first step for a decedent account is a call, not paperwork; posting before that call usually results in the paperwork being returned.

That second condition is where Mitch's track and Ian's track meet in a single record: the dispatch plan for the Chase form puts “telephone the institution” as step one, before anything is printed.

### 7.3 Field binding



```
interface FieldBinding {
  alias: string;      // the Anvil field alias
  label: string;      // human label
  item?: string;      // the item number on the printed form, e.g. "3f(1)"
  factKey?: FactKey;  // where the value comes from
  constant?: string;  // for boxes that never vary
  template?: { pattern, keys };
  required: boolean;
  format?: "text" | "usd" | "date" | "yesno";
}
```

`item` carries the number printed on the paper form, so a reviewer holding the physical document can check the mapping box by box.

`buildFill()` reads **only the verified ledger**. A quarantined fact cannot reach a filed document; there is a test asserting that the fabricated policy in the demonstration estate never appears in any payload. A field with no supporting fact is left empty and reported with the specific fact needed and where to obtain it.

## 7.4 Anvil integration

| Operation | Mechanism                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Register  | <code>createCast</code> with <code>aliasIds</code> set to our own binding aliases, so Anvil's field detection maps directly onto them. Publishes automatically, since an unpublished cast cannot be filled. |
| Reconcile | The cast query returns <code>fieldInfo</code> ; bindings are compared in both directions.                                                                                                                   |
| Fill      | <code>POST /api/v1/fill/{castEid}.pdf</code> , response written as binary.                                                                                                                                  |
| Sign      | <code>createEtchPacket</code> with a <code>castEid</code> reference and the same field-keyed payload as fill; embedded signer via <code>generateEtchSignURL</code> .                                        |
| Retrieve  | Document-group zip download.                                                                                                                                                                                |

### Why reconciliation exists at all.

Anvil's fill endpoint fails silently. A value written to a field alias the template does not have is dropped, and you receive a PDF that looks entirely correct with an empty box in the middle of it. On a probate petition that is a rejected filing and another month of the family's life. Reconciliation makes binding drift visible before anyone files, rather than after.

Registration hands Anvil our own alias list, which is what makes adding the hundredth form the same amount of work as adding the second — Ian's “thousands of forms, not five or six”.

GraphQL uses the same HTTP Basic authentication as the REST endpoint, with the API key as username and an empty password. Application errors are inspected on the response body, because GraphQL reports them with HTTP 200 — a naive status check would treat a failed mutation as success. The API key never leaves the Rust process.

## 8. Jurisdiction layer

### 8.1 California statewide

Thresholds for decedents dying on or after 1 April 2025, each verified against Judicial Council form DE-300 [Rev. April 28, 2025].

| Procedure                            | Current   | Superseded | Authority                    |
|--------------------------------------|-----------|------------|------------------------------|
| § 13100 small estate affidavit       | \$208,850 | \$184,500  | Prob. Code §§ 13100, 13101   |
| § 13151 primary residence petition   | \$750,000 | \$184,500  | AB 2016, Stats. 2024 ch. 331 |
| § 13200 real property of small value | \$69,625  | \$61,500   | Prob. Code § 13200           |
| Waiting period — §§ 13100, 13151     | 40 days   | —          | Prob. Code § 13100           |
| Waiting period — § 13200             | 6 months  | —          | Prob. Code § 13200           |

Next adjustment: 1 April 2028. Routes modelled: § 13100 affidavit, § 13151 primary-residence petition, § 13200 affidavit, spousal property petition (no dollar limit), formal probate, and DMV vehicle transfer.

#### The interaction that drives the demonstration.

§ 13100 excludes “any property included in a petition filed under Section 13151”. While the residence qualifies under § 13151 it stays outside the § 13100 computation. Cross \$750,000 and it loses § 13151 eligibility *and* falls back into the § 13100 sum — closing both economical routes at once. That cascade is what the statute says, not a contrivance, and it is why a twenty-thousand-dollar change in an appraisal can cost a family a year.

### 8.2 All 58 counties

Two kinds of county knowledge, deliberately distinguished.

| Kind        | Coverage        | Basis                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filing fees | All 58 counties | The Judicial Council schedule states that asterisked fees vary “only in the counties of Riverside, San Bernardino and San Francisco” — an exhaustive negative statement, so the correct fee is known everywhere from one citation. Riverside and San Francisco charge \$450; the other 56 charge \$435. |
| Local rules | 3 of 58         | Los Angeles, San Francisco and San Mateo were read against their published text. The remaining 55 generate a rule that says they are unverified.                                                                                                                                                        |

#### An unresearched county is a known unknown, not an absence of requirements.

Quietly defaulting 55 counties to “nothing special applies” would be exactly the confident-wrong-answer failure this architecture exists to prevent. So each generates a rule stating that its local rules have not been read, and telling the operator to check them before filing.

Researched counties carry their local forms (PRO 010, PR-5, PR-13), e-filing exclusions, examiner and tentative-ruling practice, and stale-data warnings — including that lacourt.ca.gov still serves the superseded 2022 chapter four while the operative version is served from a different host, so a scraper pointed at the obvious URL silently ingests four-year-old rules.

Switching county re-decides the estate with **no fact re-extracted**, because everything downstream is a pure function of the facts. That is the concrete demonstration that the architecture scales to three thousand counties, even though the content does not yet.

## 9. Efficiency layer

### 9.1 Correspondence

Module `correspondence.ts`. Seven reviewed templates: institution notification, institution-wide account search, foreign enquiry, IRS transcript cover, professional file request, weekly family update, and creditor notice.

- Merge fields resolve only from the verified ledger.
- Dates are spelled out for a human reader rather than left as ISO strings.
- Enclosures are computed from what the recipient actually requires — the foreign enquiry automatically lists the apostille and the certified translation.
- A letter with an unresolved required field is **refused**, not sent with a gap in it. A letter reaching a bank with a raw placeholder in it is worse than no letter at all.

#### No model writes these.

The wording is a template reviewed once; the facts are warranted; the specialist reads and presses send. That is the difference between “AI wrote your letter” and “the letter is correct” — and it is exactly the workflow Hugh described.

### 9.2 Entity resolution

Module `entities.ts`. The same asset appears under different descriptions in different documents: the will says “my accounts at Wells Fargo”, a statement says “MARGARET E HOYT, account ending 4471”, a call transcript says “her checking account”.

Matching is deterministic and evidence-bearing, across four signals: account tail, normalised institution name, holder name, and value. A differing account tail is **decisive against** a merge whatever else agrees.

| Strength    | Meaning                                   | Recommendation                      |
|-------------|-------------------------------------------|-------------------------------------|
| conclusive  | Account tail and institution both agree   | Safe to merge; record the evidence  |
| strong      | One decisive signal, or two corroborating | Confirm before merging              |
| weak        | Some agreement, nothing decisive          | Obtain the account number first     |
| conflicting | Account identifiers differ                | Do not merge — these are two assets |

#### It proposes and never decides, because the cost is asymmetric.

Merging two accounts silently drops one from the inventory. Failing to merge double-counts the estate and can push it across a procedural threshold it never actually crossed. The machine has no way to know which side it is wrong on, so anything short of conclusive goes to a human.

### 9.3 Effort accounting

Module `effort.ts`. Impact is the largest judging weight, so it should be measured rather than asserted — and measured honestly, which means being explicit that these are practice estimates of manual effort, not observations of this system running.

| Bucket    | Definition                                                          |
|-----------|---------------------------------------------------------------------|
| automated | The machine produced the output; a human need not touch it          |
| assisted  | The machine produced a draft a human reviews — faster, but not free |
| manual    | Still entirely a person's job                                       |

Review time is modelled explicitly and subtracted. Anything that inflated the automated bucket by treating review as zero would be the same category of dishonesty as an uncited threshold.

Two headline figures are published, and **the honest one is smaller**: the larger excludes speaking with the family and waiting on hold, both of which are deliberately left to people because Alix's product is human-led and that time is not meant to disappear.

The single largest line is re-checking the position when a new document arrives — ninety minutes down to ten. Today that work is either a full re-review or it is skipped.

## 10. Infrastructure

### 10.1 Provider layer

Five providers plus any OpenAI-compatible endpoint. Model identifiers, prices and capability profiles are data in `catalog.ts`, because vendors rename models constantly and a rename should be a configuration edit rather than an incident. Keys live in the Rust process; the frontend receives `has_key: bool` and nothing else.

Audited against live vendor documentation. One genuine defect was found and fixed during that audit: the `connection-test` command requested a sixteen-token response, which reasoning models spend entirely on thinking, returning empty content that the code reported as a failure — a false failure on a perfectly valid key, on the exact button the documentation tells you to press first.

### 10.2 Safety and cost control

- **Injection fencing.** Estate documents come from counterparties. Anything from a document is fenced as untrusted content with forged delimiters neutralised before a model sees it, and agents are instructed to report influence attempts as findings rather than obey them.
- **Rate limiting and a hard spend cap.** Per-provider concurrency gates, exponential backoff with jitter, immediate failure on a bad key, and a USD ceiling that aborts the run — addressing Soren's concern about spending twenty dollars every time the system runs.
- **Context budgeting.** Token-aware document selection with priority; documents that do not fit are named in the prompt rather than silently dropped.

### 10.3 Interface

Thirteen panes: Estate, Data room, Fact ledger, Investigation, Decisions, Forms, Document store, Anvil, Administration, Efficiency, Jurisdiction, Change log.

Light, quiet and serif-headed, built on the Alix palette sampled from their own deck — deliberately calm rather than dashboard-like, because it is used by someone whose parent has just died. The entire demonstration runs with no network call and therefore cannot fail on a venue's wireless.

# 11. The demonstration

Margaret Ellen Hoyt died on 4 January 2026 in San Mateo County. Her daughter Claire is executor. The estate is synthetic; the law applied to it is not.

| Time | Beat            | What is shown                                                                                                                                                                                         |
|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 | The problem     | Six procedures, 58 counties, and an LLM that will hallucinate a threshold                                                                                                                             |
| 0:20 | Extraction      | Seven documents ingested; sixteen facts proposed                                                                                                                                                      |
| 0:40 | The fabrication | One claimed \$50,000 policy is quarantined — the quote exists in no document, and it never reaches the rules engine                                                                                   |
| 1:10 | Rules as data   | Exclusions applied, \$148,600 personal property derived, DE-310 selected against the real \$750,000 cap, every figure cited                                                                           |
| 1:50 | The flip        | A supplemental appraisal at \$760,000 arrives. Two decision points re-evaluated, two flipped, two provably skipped. Both economical routes close at once; three weeks becomes nine to eighteen months |
| 2:30 | Offshore        | Schedule B, Form 5471 and a Pictet wire raise critical leads; distribution is refused                                                                                                                 |
| 2:50 | Governance      | Irreversible actions gated; missing facts named with where to get them                                                                                                                                |

The demonstration is also expressed as a test file. If `ca-probate.test.ts` goes red, the demonstration is broken — which is a considerably better alarm than discovering it on stage.

## 11.1 Test inventory

| Suite              | Covers                                                     | Tests     |
|--------------------|------------------------------------------------------------|-----------|
| verify + hardening | Quote verification, fencing, limiter, budgeting            | inherited |
| engine             | Ledger, derivation, rules, reactor, gate, gaps             | 17        |
| ca-probate         | The demonstration end to end, with real thresholds         | 17        |
| ca-counties        | 58 counties, fee variance, overlays, rule corrections      | 15        |
| anvil              | Field binding, provenance, reconciliation, drift detection | 13        |
| formstore          | Composite identity, resolution, ambiguity, dispatch        | 15        |
| investigation      | Leads, deadlines, task graph, distribution gate            | 20        |
| efficiency         | Correspondence, entity resolution, effort model            | 18        |
| Rust               | PDF, DOCX, XLSX extraction against real fixtures; base64   | 11        |

**171 TypeScript tests and 11 Rust tests, all passing.**

## 12. Honest limitations

Stated plainly and up front, because a limitation volunteered is a strength and one extracted under questioning is a wound.

| Limitation               | Detail                                                                                                                                                                                              |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No live model call       | Neither this codebase nor its parent has ever made one. Extraction is stubbed with hand-written candidates in the identical FactCandidate shape, so wiring it is a small job — but it is not wired. |
| No live Anvil call       | The shapes come from documentation. The createCast file encoding and the exact etch input types are the least certain parts.                                                                        |
| 3 of 58 counties         | Local rules read for Los Angeles, San Francisco and San Mateo. The other 55 are flagged unverified, not defaulted.                                                                                  |
| Heggstad / § 850         | Not modelled — and it is the actual remedy for the unfunded trust the demonstration is framed around.                                                                                               |
| Liabilities              | Not in the ledger. Assets only; creditor work exists as tasks and deadlines.                                                                                                                        |
| Foreign jurisdictions    | No rule pack. Ancillary probate, resealing and letters rogatory appear as lead actions rather than as rules.                                                                                        |
| Investigation log        | Derived rather than persisted — recomputed each run instead of append-only with timestamps, which is what defensibility actually wants.                                                             |
| Track 2 (voice)          | Designed as an adapter over the same fact pipeline; not built.                                                                                                                                      |
| Effort figures           | Practice estimates of manual effort, not measurements of this system.                                                                                                                               |
| What verification proves | That a quotation exists, not that the conclusion follows. It eliminates fabricated evidence; it does not eliminate bad reasoning from real evidence.                                                |

### This is not legal advice.

It is analysis support that a human owns, reviews and signs — which is also Alix's own stated product position. The system is built to make a specialist faster and harder to mislead, not to replace their judgement.

### 12.1 What comes next

| Priority | Work                                                              | Effort       |
|----------|-------------------------------------------------------------------|--------------|
| 1        | Wire live extraction; run once and record the output as a fixture | 1–2 hours    |
| 2        | Register DE-310 with Anvil and complete one live fill             | 45 minutes   |
| 3        | Model the Heggstad / § 850 petition                               | 30 minutes   |
| 4        | Add liabilities and creditor claims to the ledger                 | 2 hours      |
| 5        | Persist the investigation log as an append-only record            | 2 hours      |
| 6        | Extraction council with tiered escalation and conflict surfacing  | 3 hours      |
| 7        | Foreign jurisdiction rule pack, starting with Switzerland         | half a day   |
| 8        | Track 2 voice adapter emitting the same FactCandidate shape       | several days |